Leakage-aware inverse identifiability of black-hole hair from leading-eikonal observables

Preliminary AI-for-science study

July 26, 2026

Abstract

We study inverse identifiability of analytic black-hole hair parameters from leading-eikonal ringdown and geodesic observables. Unlike a forward calculation, the inverse problem asks whether distinct parameter values remain distinguishable after compression, noise, and physically disjoint validation. We therefore use grouped physical splits, fit principal-component models only on clean training folds, and compare machine-learning error with a numerical Jacobian analysis. Across 1,577 valid dense Kiselev systems and five grouped folds, interpolation gives $R^2(k) = 0.992 \pm 0.001$ and $R^2(w_q) = 0.994 \pm 0.002$, while analytic sensitivity reveals a local rank loss near $k \simeq 0$. Smooth synthetic geodesic proxies reduce the mean absolute w_q error from 0.0225 to 0.0143, increase the median minimum singular value near $k = 0$ by approximately $12\times$, and improve the median condition number by approximately $2.13\times$. These proxies are not physical ray-tracing results. Finally, 3,337 samples from the preferred GW150914 `C01:Mixed` posterior are propagated through an approximate Kerr $(2, 2, 0)$ fitting formula to provide an observational-scale toy tolerance, not a hair detection or modified-gravity posterior. The study remains limited to static leading-eikonal formulas, synthetic observables, and a GR posterior-scale comparison without detector covariance or a non-GR likelihood.

1 Introduction

Analytic forward formulas specify how black-hole parameters modify photon-orbit quantities and eikonal quasinormal-mode (QNM) frequencies. Forward predictability, however, does not guarantee inverse identifiability. A parameter may be globally learnable by interpolation yet locally degenerate, or may appear recoverable only because repeated observations of the same physical system cross a random-row train/test split. Observational relevance is a third question: even an identifiable synthetic parameter may produce shifts smaller than current posterior uncertainty.

This work separates these questions. First, leakage-aware grouped validation tests synthetic inverse recovery. Second, a Jacobian singular-value analysis locates analytic degeneracy. Third, replaceable synthetic geodesic proxies test whether radially distinct observables improve conditioning without adding ML complexity. Finally, a public GW150914 remnant posterior provides a real posterior-scale comparison. It is not reanalyzed with a non-GR waveform.

2 Physics setup

We use geometrized units $G = c = 1$, with $M = 1$ unless mass is varied explicitly. The Schwarzschild reference quantities are

$$r_0 = 3M, \qquad \Omega_0 = \lambda_0 = \frac{1}{3\sqrt{3}M}. \tag{1}$$

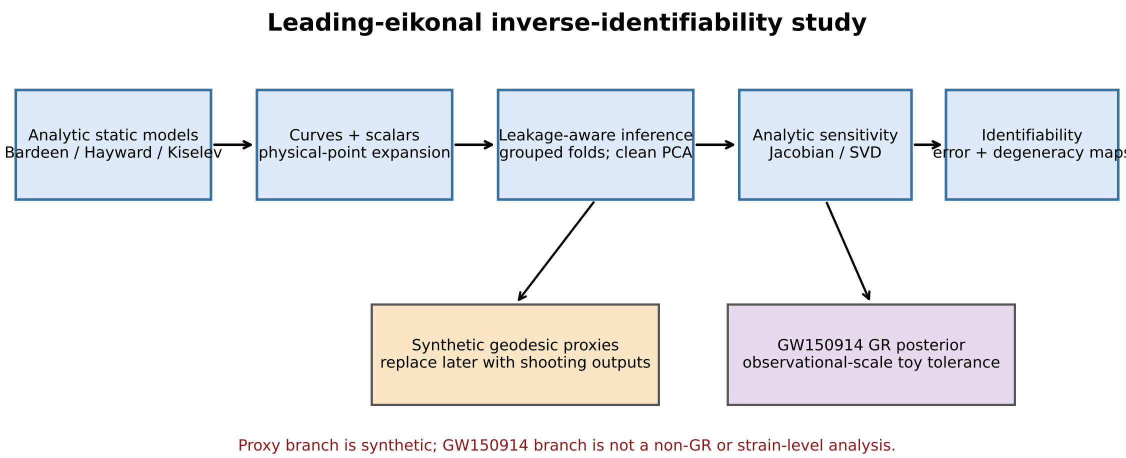

Figure 1: Study design. Analytic leading-eikonal simulation is followed by physical-group validation and Jacobian sensitivity; a replaceable synthetic-geodesic branch tests additional information, while the GW150914 branch supplies only a GR posterior-scale toy tolerance.

The leading-eikonal QNM relation and illustrative damped curve are

$$\omega_{\rm QNM} = \ell\Omega - i(n + \tfrac{1}{2})\lambda, \qquad \Psi(t) = e^{-(n+1/2)\lambda t}\cos(\ell\Omega t). \tag{2}$$

For Bardeen,

$$\frac{\Omega}{\Omega_0} = 1 + \frac{q^2}{6M^2}, \quad \frac{\lambda}{\lambda_0} = 1 - \frac{q^2}{9M^2}, \quad \delta r = -\frac{5q^2}{6M}. \tag{3}$$

For Hayward,

$$\frac{\Omega}{\Omega_0} = 1 + \frac{q^3}{27M^3}, \quad \frac{\lambda}{\lambda_0} = 1 - \frac{2q^3}{27M^3}, \quad \delta r = -\frac{2q^3}{9M^2}. \tag{4}$$

For Kiselev,

$$\frac{\Omega}{\Omega_0} = 1 - \frac{3k}{2(3M)^{1+3w_q}}, \tag{5}$$

$$\frac{\lambda}{\lambda_0} = 1 + \frac{[3w_q(1+w_q) - 2]k}{4\,3^{3w_q} M^{1+3w_q}}, \tag{6}$$

$$\delta r = \frac{3k(1+w_q)}{2(3M)^{3w_q}}. \tag{7}$$

These are static, leading-order expressions rather than dedicated gravitational perturbation or low-ℓ QNM calculations. The synthetic data test inference only within the assumed formula families.

3 Machine learning and leakage-aware methodology

Each non-Schwarzschild physical parameter point is expanded over all configured (ℓ, n) combinations. Physical groups exclude ℓ and n: Bardeen and Hayward use (family, M, q), and Kiselev uses (family, M, k, w_q). All repeated modes from a physical system remain in one train, validation, or test partition.

A random-row audit demonstrated that correlated repetitions can substantially inflate sparse Kiselev recovery. Subsequent analyses use five-fold grouped validation. Principal component analysis (PCA), whenever used, is fitted only on clean training curves inside each fold; validation and test curves are transformed without refitting. The repeated grouped study reports fold means and standard deviations rather than a single favorable split.

4 Kiselev identifiability and analytic degeneracy

The dense study contains 1,577 valid physical Kiselev systems after rejecting non-positive or non-finite leading-order points. Five grouped folds give

$$R^2(k) = 0.992 \pm 0.001, \qquad R^2(w_q) = 0.994 \pm 0.002. \tag{8}$$

Thus, with dense coverage, the parameters are globally learnable by interpolation. This does not imply uniform local identifiability.

For the observable vector

$$\mathbf{O} = (\Omega, \lambda, \omega_R, \gamma, \delta r), \tag{9}$$

we compute the numerical Jacobian

$$J_{ij} = \frac{\partial O_i}{\partial \theta_j}, \qquad \boldsymbol{\theta} = (k, w_q). \tag{10}$$

Large condition number and small minimum singular value identify locally ill-conditioned regions. The strongest degeneracy lies near $k \simeq 0$, where all Kiselev corrections vanish with k. Dense grouped-CV w_q error is moderately correlated with the Jacobian condition number, so analytic sensitivity explains a meaningful part of the spatial ML error pattern.

5 Feature realism

The illustrative waveform contains no information independent of Ω and λ once ℓ and n are specified: $\omega_R = \ell\Omega$ and $\gamma = (n + 1/2)\lambda$. PCA coefficients therefore compress a deterministic re-expression of those quantities. PCA-only recovery is weaker, and adding PCA to Ω/λ does not improve grouped performance. All current scalar observables perform best because δr supplies additional geometric information not contained in the damped waveform.

6 Independent geodesic-observable extension

We next add an impact-parameter proxy, apparent screen-coordinate proxy, propagation-time-delay proxy, and redshift-curve proxy. They are smooth, metric-based placeholders designed for later replacement by direct/secondary null-geodesic shooting outputs. **They are synthetic geodesic proxies, not physical ray-tracing predictions.**

The mean absolute w_q error decreases from 0.0225 to 0.0143. Near $k = 0$, the median minimum singular value of the enlarged observable Jacobian improves by approximately 12×, and the median condition number improves by approximately 2.13×. Different radial functions therefore reduce the near-degeneracy for small nonzero k. The exact $k = 0$ rank loss remains fundamental: if the perturbation amplitude vanishes, the metric has no w_q dependence.

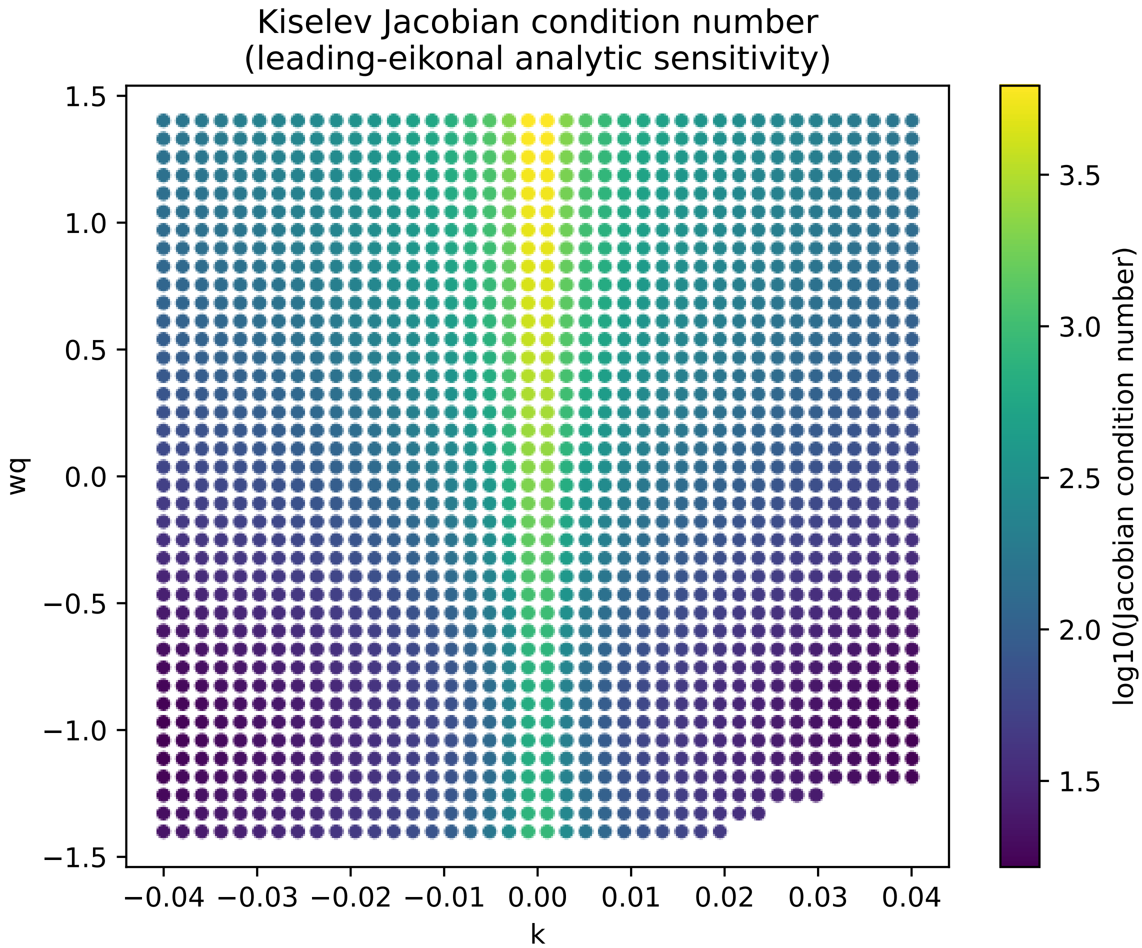

Figure 2: Dense Kiselev Jacobian condition number. Large values mark local ill-conditioning; absolute values depend on observable scaling.

Table 1: Five-fold grouped-CV results. The geodesic values are synthetic proxies, not physical ray-tracing observables.

Feature set	$R^2(k)$	$R^2(w_q)$
Ω/λ only	0.9393	0.9680
All current scalars	0.9957	0.9974
Geodesic proxies only	0.9953	0.9979
Current scalars + proxies	**0.9972**	**0.9985**

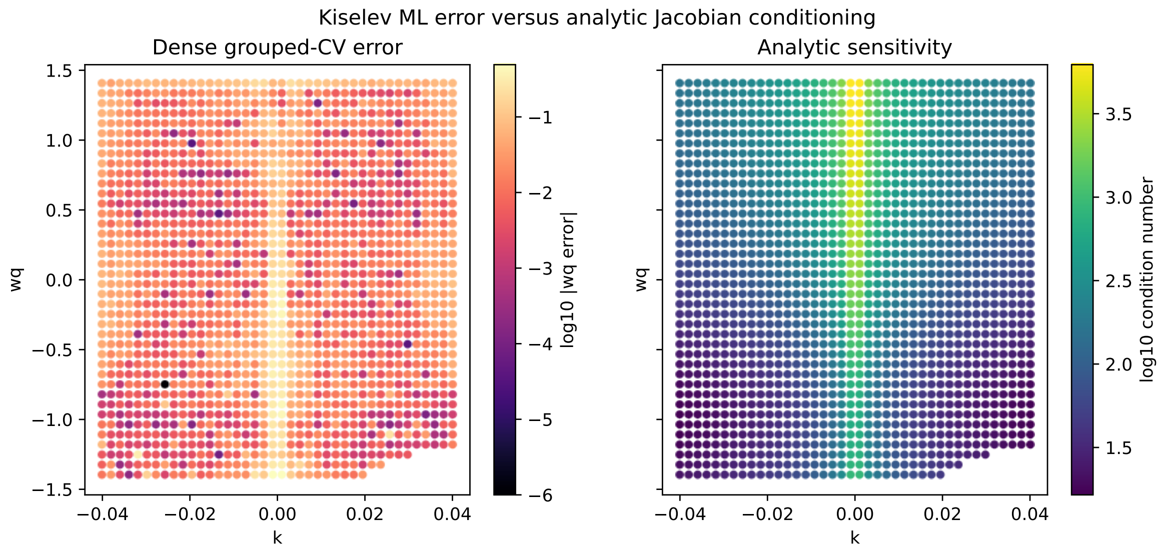

Figure 3: Dense grouped-CV w_q error compared with analytic Jacobian sensitivity. Their spatial association supports a local identifiability interpretation rather than a pure estimator-failure interpretation.

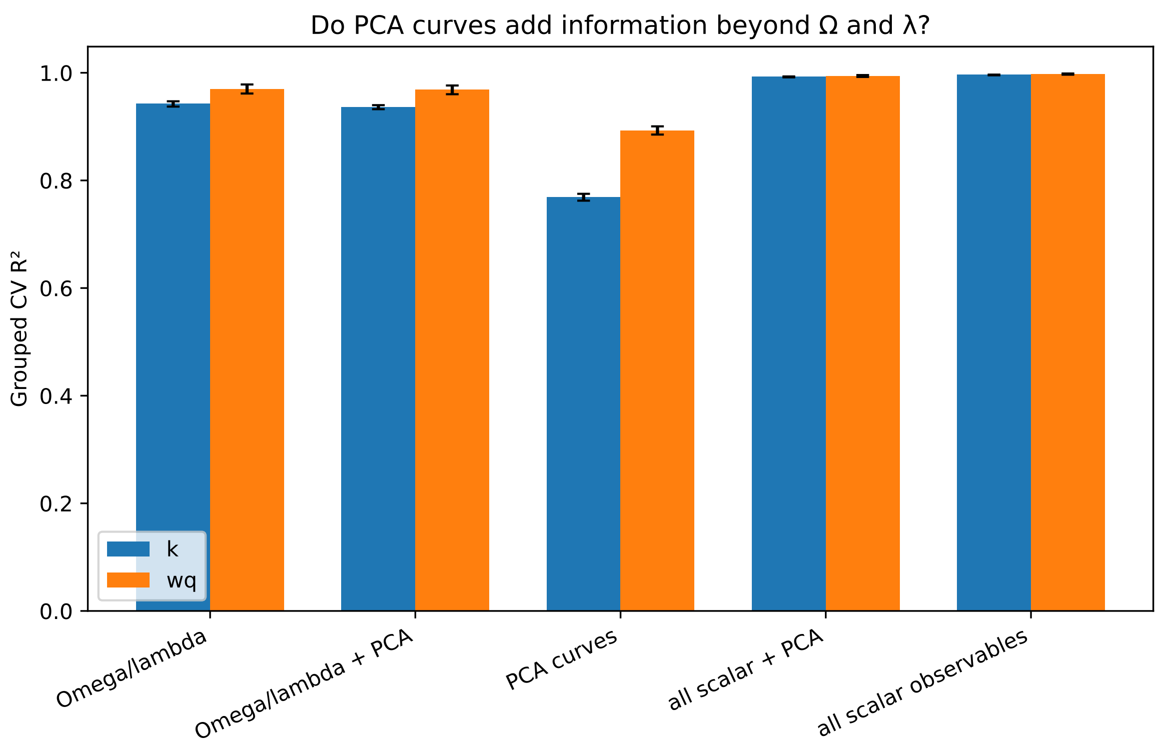

Figure 4: Five-fold grouped feature comparison for dense Kiselev inference. PCA waveform features do not create independent physics information beyond the quantities used to generate the waveform.

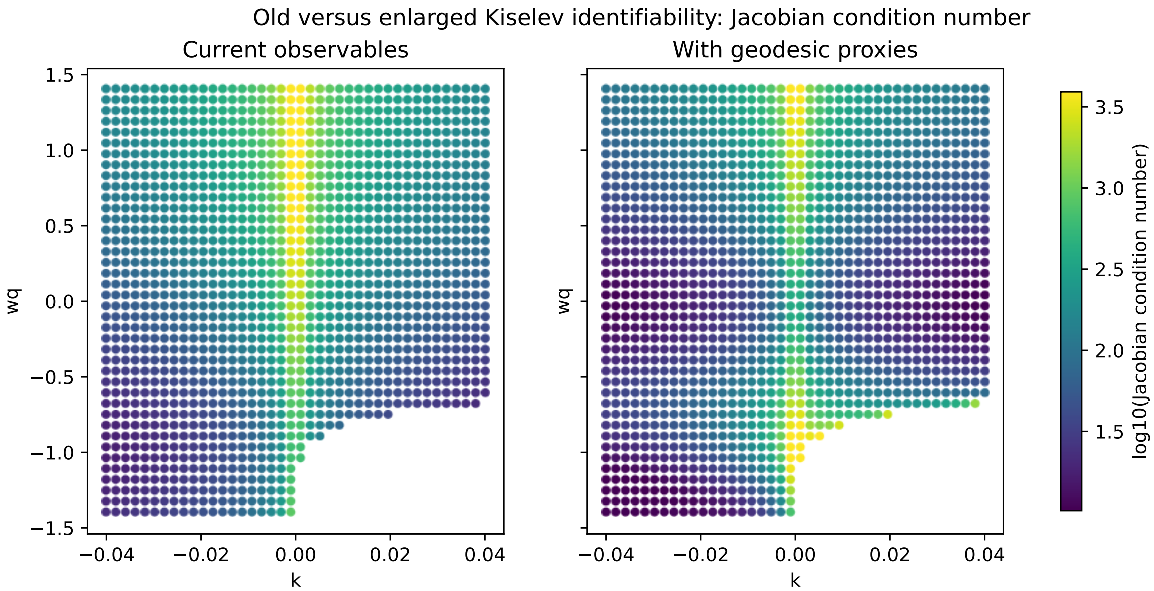

Figure 5: Old versus enlarged-observable condition-number maps. Improvements use synthetic geodesic proxies and are not ray-tracing forecasts.

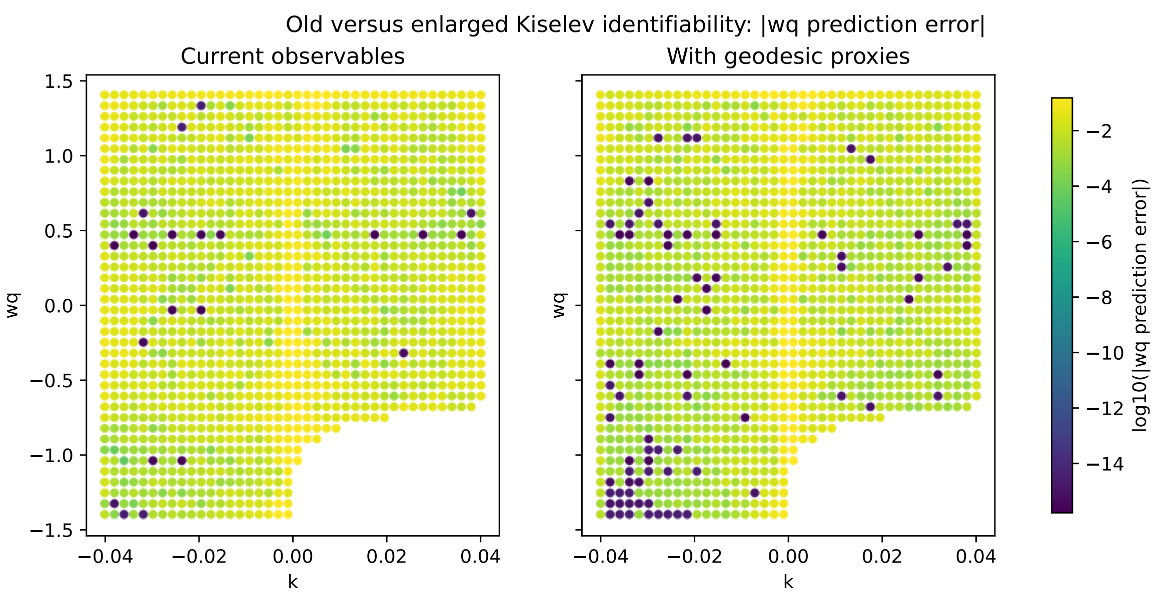

Figure 6: Old versus enlarged-observable grouped-CV w_q error. The right panel uses synthetic geodesic proxies, not physical observations.

7 Public GW150914 posterior-scale comparison

We use 3,337 samples from the preferred GW150914 `C01:Mixed` public posterior. Detected fields include source- and detector-frame final mass, final spin, redshift, luminosity distance, component masses, and mass ratio. Because a usable `qnm` backend was not available, the dominant Kerr $(2, 2, 0)$ mode is computed with a clearly labeled approximate fitting formula. With $GM_{\odot}/c^3 = 4.925490947 \times 10^{-6}$ s,

$$f_{\rm RD} = \frac{M\omega_R}{2\pi M_z}, \qquad \tau_{\rm RD} = \frac{M_z}{|M\omega_I|}. \tag{11}$$

The propagated posterior has median $f_{\rm RD} = 252.52$ Hz with 90% interval 245.01–260.66 Hz, and median $\tau_{\rm RD} = 4.065$ ms with interval 3.809–4.380 ms. The dimensionless medians are $M\omega_R = 0.5285$ and $M\omega_I = -0.0820$.

Treating the posterior half-widths only as a toy tolerance, the complete configured Bardeen range $q/M \leq 0.25$, the complete Hayward range $q/M \leq 0.35$, and approximately 60.9% of the valid configured Kiselev grid are allowed under the toy tolerance. These are not LIGO exclusions, a hair detection, a modified-gravity posterior bound, or a strain-level analysis. They propagate uncertainty from a GR remnant posterior and place the synthetic shifts on an observational scale.

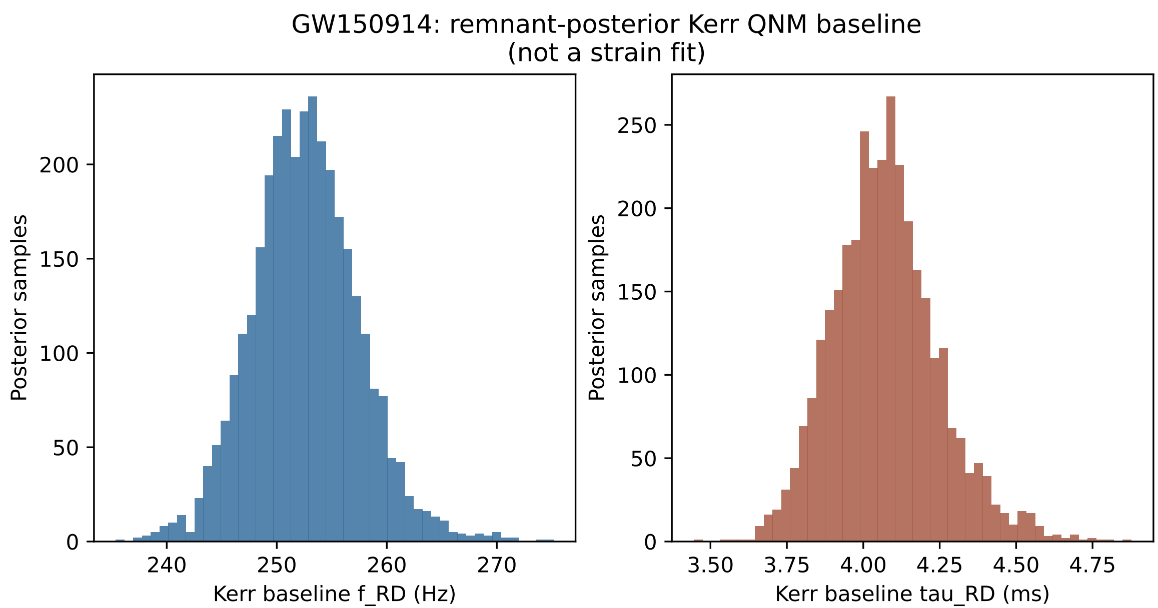

Figure 7: GW150914 remnant-posterior propagation through the approximate Kerr $(2, 2, 0)$ baseline. This is a posterior-scale comparison, not a non-Kerr inference.

8 Discussion

Machine learning establishes whether the assumed analytic inverse map can be interpolated under physically disjoint validation. Analytic sensitivity explains where that map becomes locally ill-conditioned and why dense average scores can coexist with a rank-deficient limit. The feature audit shows that deterministic waveform compression cannot manufacture independent information. The synthetic geodesic extension suggests that observables probing different radial structure can reduce near-degeneracy, but this conclusion must be retested with real shooting calculations.

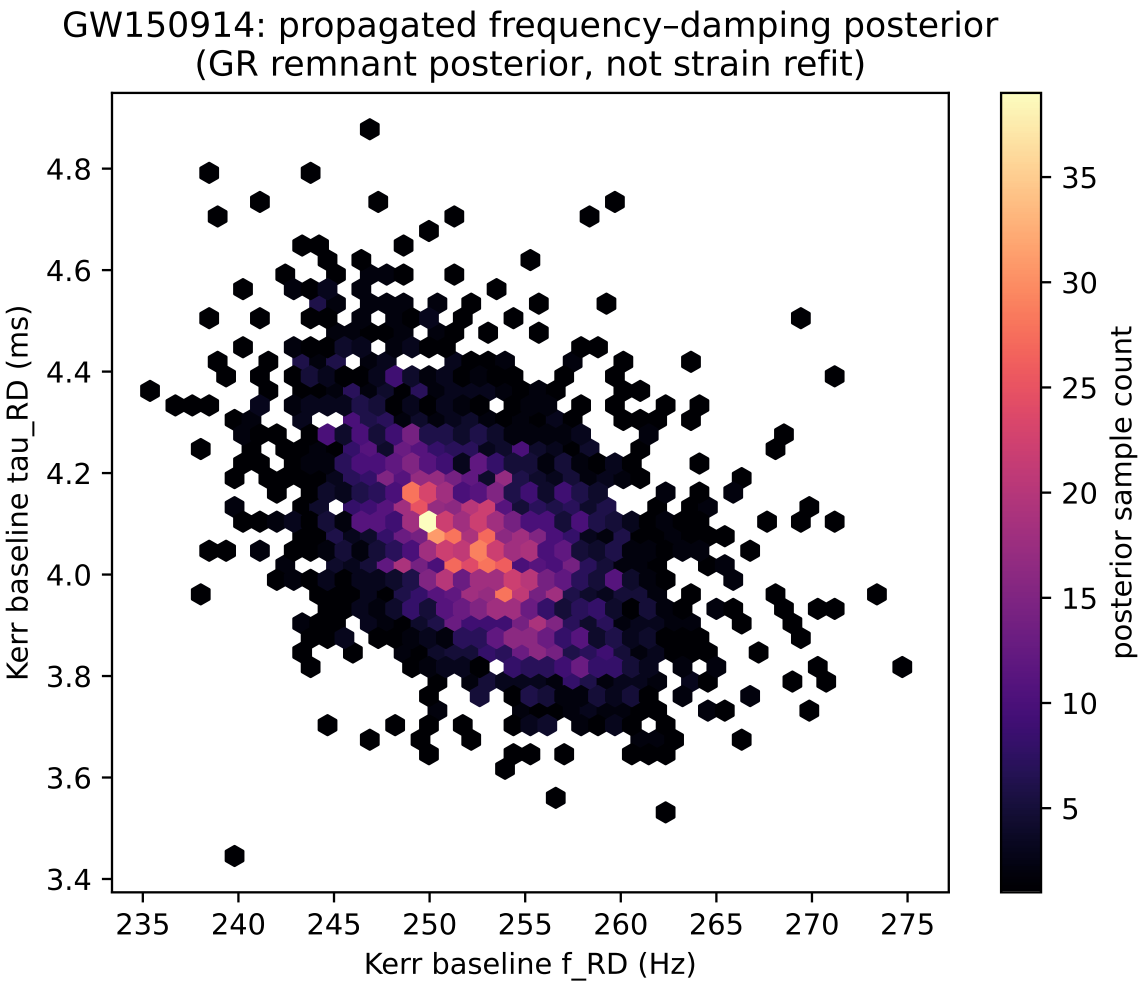

Figure 8: Joint GW150914 frequency–damping posterior implied by the GR remnant posterior and approximate Kerr fit; no strain-level ringdown refit is performed.

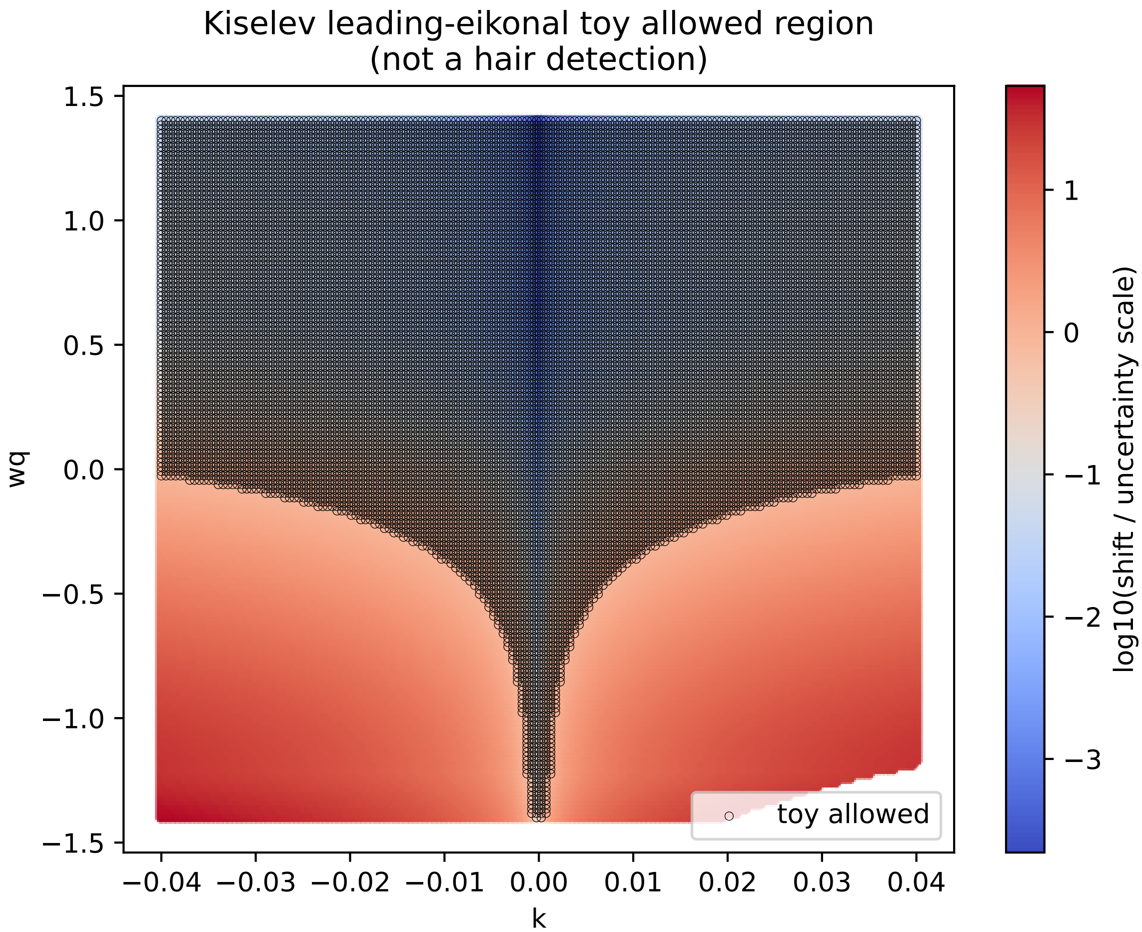

Figure 9: Kiselev region allowed under the toy leading-eikonal tolerance defined by the GW150914 posterior half-widths. It is not an observational modified-gravity bound.

GW150914 currently contributes an uncertainty scale only; it does not supply a likelihood for the hair parameters.

9 Limitations

- The physics uses static, leading-eikonal toy hair formulas rather than dedicated gravitational perturbation calculations.
- The independent geodesic quantities are synthetic proxies, not physical ray-tracing or null-geodesic-shooting results.
- The Kerr baseline uses an approximate dominant-mode fit because a usable `qnm` backend was unavailable.
- No detector covariance, calibration uncertainty, selection effect, or ringdown-start-time uncertainty is propagated.
- No non-GR likelihood, strain-level inference, or dedicated QNM validation is performed.

10 Next steps

The immediate priority is to replace the proxy table with converged direct and secondary null-geodesic-shooting outputs, then rerun the enlarged Jacobian and grouped-CV audit with an observational covariance model. Subsequent work should install and validate the `qnm` package or trusted QNM tables, analyze multiple public events, calibrate predictive uncertainty, include detector-informed covariance, and eventually construct a consistent non-GR likelihood.

11 Conclusion

Dense grouped validation shows that Kiselev (k, w_q) is globally learnable by interpolation within the analytic leading-eikonal family, while Jacobian sensitivity reveals a local degeneracy near $k = 0$. Synthetic geodesic proxies reduce the local error and improve conditioning, suggesting that genuinely independent geodesic observables are a promising next diagnostic; these proxy gains are not physical ray-tracing results. The exact $k = 0$ rank loss remains fundamental because the metric has no w_q dependence when the perturbation vanishes. The GW150914 exercise places the shifts on a real posterior scale but remains a toy-tolerance comparison, not a hair detection.